Name:

Matriculation number:

/ 20

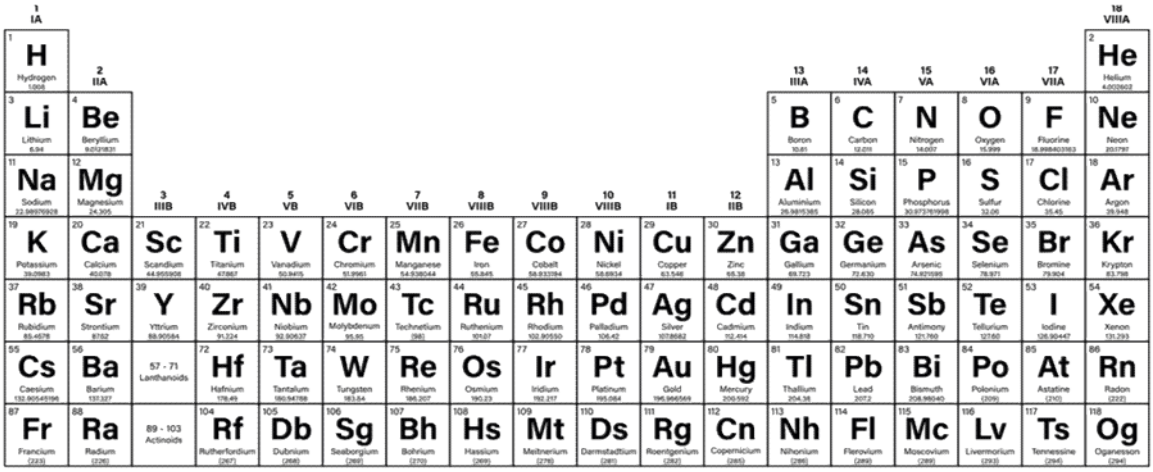
Instructions for CM2021 Mid-Term Test

1. This test paper contains Twenty (20) multiple-choice questions

2. Answer ALL questions (1 mark each)

3. Time allowed: 1 hour

4. Do not turn over the test paper until told to do so



1) Which of the following is a coordination compound?

a. Platinum
b. Fe^{2+} ion
c. Cisplatin
d. Sulfuric acid

2) The central metal atom in $[Pt(NH_3)_2Cl_2]$ is:

a. Pt^{2+}
b. Pt^{0}
c. Pt^{4+}
d. None of the above

3) Which of the following complexes can exhibit both geometrical and optical isomerism?

a. $[Pt(NH_3)_2Cl_2]$
b. $[Cr(en)_3]^{3+}$
c. $[Co(en)_2Cl_2]^{+}$
d. Both b and c are correct

4) Which of the following compound shows only linkage isomerism?

a. $[Co(NH_3)_5(SCN)](NO_2)$
b. $[Co(ONO)(NH_3)_3]NO_2$
c. $[Co(en)_3]Cl_2$
d. $[Co(NO_2)Br(NH_3)_4]Cl$

5) How many isomers are possible for the complex $[Pt(NH_3)(H_2O)(SCN)(NO_2)]$

a. 4
b. 8
c. 12
d. None of the above

6) Which of the following is the correct name for $[Cu(H_2O)_6][Pt(CN)_6]$:

a. Hexaaquacopper(II) hexacyanoplatinate(II)
b. Hexaaquacopper(II) hexacyanoplatinate(IV)
c. Hexaaquacopper(I) hexacyanoplatinate(II)
d. Hexaaquacopper(II) hexacyanoplatinum(IV)

7) The name of $[Fe(CN)_2((CH_3)_3P)_2]$ is:

a. Dicyanobis(trimethylphosphine)iron(II)
b. Dicyanodi(trimethylphosphine)iron(II)
c. Dicyanobis(triphenylphosphine)iron(IV)
d. Dicyanotris(methyl)bis(phosphine)iron(IV)

8) Which of the following gives the highest number of isomers?

a. $[Co(NH_3)_4Cl_2]$
b. $[Ni(en)(NH_3)_4]^{2+}$
c. $[Ni(C_2O_4)(en)_2]$
d. $[Cr(SCN)_2(NH_3)_4]^{4+}$

9) What is the name for $[Zn(OH)_4]^{2-}$?

a. Tetrahydroxozincate(II)
b. Tetrahydroxozincate(IV)
c. Tetrahydroxyzincate(II)
d. Zinc tetrahydroxide

10) What is the name for $Na_2[Fe(C_2O_4)_2(H_2O)_2]$?

a. Sodium diaquabis(oxalato)ferrate(III)
b. Sodium diaquabis(oxalato)ferrate(II)
c. Sodium oxalatoferrate(III)
d. Sodium diaquabis(oxalate)iron(III)

11) Which of the following metal atomic orbitals are considered in crystal field theory?

a. d only
b. p and d
c. s and d
d. s, p, and d

12) What causes the metal d-orbitals to lose their degeneracy in the crystal field theory?

a. Spherical field
b. Point charges
c. Repulsion between metal's d and p orbitals
d. None of the above

13) In $[V(CO)_6]^{5+}$, the V-C bond possesses:

a. π character only
b. σ character only
c. ionic character only
d. both π and σ character

14) Which of the orbital will be most stablized if two axial ligands (i.e., along the z-axis) are removed?

a. d_{xz}
b. d_{yz}
c. d_z^2
d. d_{xy}

15) What is the electronic arrangement of metal atom/ion in octahedral complex with d^4 configuration? Assume pairing energy > Δ_o

a. $t_{2g}^4 e_g^0$
b. $e_g^4 t_{2g}^0$
c. $t_{2g}^3 e_g^1$
d. $e_g^2 t_{2g}^2$

16) Which of the following is true when an octahedral metal complex undergoes axial elongation?

a. d_{xy} orbital has the lowest energy
b. d_{x2-y2} orbital has the highest energy
c. Metal d-orbitals with lobes pointing towards the Z axis are destablized
d. All of the above

17) Which of the following on ammine ligand is true?

a. σ-donor, π-acceptor
b. σ-donor, π-donor
c. σ-acceptor, π-acceptor
d. None of the above

18) Identify the first-row transition metal M that satisfies the requirements given (more than one answer may be possible).
$[M(H_2O)_6]^{2+}$ having LFSE = $0.6\Delta_o$.

a. Cr
b. Cu
c. Co
d. Both (a) and (b) are correct

19) What is the CFSE of a low-spin d^6 metal complex?

a. $2.4\Delta_o$ - 3P
b. $2.4\Delta_o$ - 2P
c. $2.4\Delta_o$ - P
d. $2.4\Delta_o$

20) Which effect/theory could account for the presence of two absorption peaks for a d^1 metal complex?

a. Jahn-Teller effect
b. Metal-to-ligand pi back bonding
c. Ligand-to-metal pi bonding
d. None of the above

End of test paper.